Virginia Commonwealth University — Fall 2025
School of Engineering

EGRE 245– Course Syllabus
Engineering Programming (4 Credits)

Instructor Contact: Mark Schwitzerlett
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Course Meeting: Monday and Wednesday 4:00 - 5:15 pm Engineering Building East E2214

Lab Meeting: Tuesday 2:00 - 3:50 pm Engineering Building East E1239
OR
Thursday 2:00 - 3:50 pm Engineering Building East E1239

Student Hours: Monday and Wednesday 3:00 - 3:50 pm
Tuesday and Thursday 12:30 - 1:30 pm
or by appointment

Prerequisite: MATH 151 (with a minimum grade of C)

Textbook:

- Online Zybook **(required)**
  1. Sign up here
  2. Enter the code **VCUEGRE245Fall2024**
  3. Click Subscribe
- *The C Programming Language*, 2nd Edition, Brian W. Kernigan and Dennis M. Ritchie, Prentice Hall 1988 **(optional)**

Course Description: This is the first course in structured programming using the C language. Topics covered include the concepts and practice of structured programming, problem solving, and top-down design of algorithms. Basic C syntax, control structures, and arrays and data structures will be presented. In order to master the material being covered during the semester, programming problems will be assigned in addition to the presented lecture material.

Course Objectives:
Upon successful completion of this course, the student will be able to:

A. understand the concepts and practice of structured programming;
B. understand problem solving and top down design of algorithms;
C. program using basic C syntax;

D. understand the use control structures, functions and arrays;
E. be able to develop and debug basic computer programs.

Reading Assignments and Homework: These assignments will be completed in the required Zybook for the course through links in Canvas.

Programming Assignments:

- Clearly, programming is a major portion of this course. As such, programs will count 35% of the final grade.
- **Late programs will NOT be accepted unless arrangement is made PRIOR to the due date. Please speak to the instructor or the TAs if you are having problems.**
- All programs are due at midnight on the day specified.
- There will be approximately 9 programs for you to write.
  - Tentative due dates for the assignments are listed on the attached course schedule.
  - The programming assignments will increase in size and complexity as the course progresses.
  - It is to your advantage to hand in partial programs if you cannot complete a particular assignment.
  - All assignments will be submitted via Canvas.
    - If your solution is a single file (such as a single "program1.c" file) then you can submit just the one file.
    - If your solution consists of multiple files (such as a "program3.h" and "program3.c" file), you must zip them up together and submit a single .zip file.

TopHat: Students are expected to attend class every day. TopHat will be used to record attendance and understanding of class material. There will be multiple TopHat questions during each class session. TopHat will be graded 50% for participation and 50% for correctness.

Quizzes: Approximately seven in-class quizzes will be given in the semester. The lowest two in-class quiz grades will be dropped and all remaining in-class quiz grades will be averaged at the end of the semester. All in-class quizzes are considered pledged. **Quizzes will not be announced beforehand. In-class quizzes that are missed for anything other than serious emergency situations cannot be made up.**

Lab Exercises:

1. **Attendance is mandatory**. Attendance will be recorded using TopHat in Lab as well.
2. Outside of situations not under your control, lab assignments must be completed during the **lab session**.
   - If you arrive at the lab on time and cannot complete the assignment by the time the lab finishes, then the lab instructor may give you permission to submit the assignment within 24 hours of the end of the lab.

  - This extension is not guaranteed – you must receive permission from the lab instructor, and it will never be for more than 24 hours.

3. Your work must be "graded" by the lab instructor or TA. The instructor will notify you each lab as to what they expect from you during that session.
4. You will work in pairs during lab sessions. Partners will be assigned and will probably change during the semester. You and your partner will both receive the same grade for the lab session (assuming you both attend the lab and turn in the same code for the lab). Note that working in pairs DOES NOT apply to the class programming assignments.
5. You may leave early from a lab if you have completed the work to the satisfaction of the lab instructor. Generally this will mean you have completed that day's work plus had it checked out or graded by the instructor.
6. You should only work on lab assignments while attending lab sessions.
7. You may not switch lab sessions without prior permission from both lab and course instructor.
8. You may bring and use your own laptop computer.

Midterm Exam: The midterm exam will be 1 hour and 15 minutes in length and will be given in class. The approximate date for this midterm exam tests is given in the class schedule, but is subject to change. The exact date for the midterm exam will be announced in class at least two lectures before it occurs. **A midterm exam that is missed for anything other than serious emergency situations cannot be made up. An excused absence from the midterm exam will require some form of written documentation from an "official" 3rd party.**

Final Exam: **Wednesday, December 10, 4:00 - 6:50 pm**

Grade Calculation:

Category	Weight
Reading Assignments	5%
Homework	5%
TopHat	5%
Quizzes	10%
Lab Exercises	15%
Programming Assignments	35%
Midterm Exam	15%
Cumulative Final Exam	20%

Course Average	Letter Grade
90 – 100	A
80 – 89	B
70 – 79	C
60 – 69	D
0 – 59	F

Grading of Programming Assignments:

- 0% for assignments not submitted by midnight on the due date
- Base Grade for Compilation and Running
  - 25% Program will not compile
  - 50% Program compiles, but will not run
  - 75% Program runs, but with incorrect results
  - 100% Program runs with correct results and meets all specifications
- Well written Program
  - 40% of base grade may be forfieted for poor program design
  - 30% of base grade may be forfieted for style and documentation errors
  - 20% of base grade may be forfieted for lack of user-friendliness and/or incorrect output format
  - 10% of base grade may be forfieted for lack of robustenss and portability

Therefore, a program that runs perfectly with poor formatting and no documentation would receive 70% of the possible points. Programs that do not include the standard class header in all source code files will receive a grade of 0 for the style and documentation component!

Unless otherwise stated in the homework problem itself, all programs must compile and execute correctly from the command line (i.e., not in the "Visual Studio environment"). *Programs that do not compile or execute correctly from the command line will be scored as not compiling or executing regardless of whether they do so in Visual Studio.*

Important Dates:

T	Aug 19	First day of class
M	Aug 25	Last day of add/drop
T - M	Aug 26 - Sep 15	Progress reports
F	Aug 29	Deadline to notify intent to observe religious holidays
M	Sep 1	Labor Day – No class
F	Oct 17	Reading Day – No class & Midterm grades due
F	Oct 31	Last day to withdraw and Last day to request the pass/fail option
T	Nov 4	Election Day – No class
M - F	Nov 24 - 28	Fall Break – No class
M	Dec 8	Last day of classes
F	Dec 12	Final exam 12:30 - 3:20 pm

VCU Honor System:
The VCU Honor System policy describes the responsibilities of students, faculty and administration in upholding academic integrity. According to this policy, "Members of the academic community are required to conduct themselves in accordance with the highest standards of academic honesty, ethics and integrity at all times." Students are expected to read the policy in full and learn about requirements here.

Course Standards for Cheating and Plagiarism:
It is important that the work you hand in is your own. It is fairly easy to copy someone else's program (especially as the due date approaches). For you to learn the material in order to perform well on the tests and subsequent programs, it is imperative that you understand everything that is in your program. Obtaining verbal assistance on programs is allowed. However, programming is a learn-by-doing skill. The homework assignments are key tools for learning the material in the course. **Doing the programming assignments as "group projects" is strictly prohibited and will invariably lead to you performing poorly in the course.** Make sure that the work you hand in is truly yours. Three simple guidelines to ensure this are:

1. The "form of expression" must be original.
2. Understand everything in your program.
3. Never copy files from someone else, never give files to someone else, never examine someone else's code, never show someone else your code.

The internet is a great resource for finding alternate explanations or examples for programming concepts that may help you better understand the material and you are allowed, and encouraged to utilize it. However, asking for specific help that is targeted towards a programming assignment on blogs or forums is considered cheating and will not be tolerated. The TA's and the instructor are much better sources of assistance anyway. Code that is obtained from outside sources, such as the internet, and used in a homework submission MUST be properly cited in the code comments or it will be considered plagiarism. Homework submissions that include a significant amount of (properly cited) copied code may receive a lower grade. Furthermore, the use of AI is ***STRICTLY PROHIBITED***.

If a student hands in work that violates the standards outlined above, it will be dealt with it appropriately. The TAs will be using Stanford University's Moss (for Measure Of Software Similarity) program to detect software plagiarism. You are reminded that all homework is pledged to be your own work as described above and it is an honor code violation to misrepresent work that is not yours. Programs that violate these standards will be referred to the Honor System for action.

Course Policy for Grading Issues: If you notice a problem with the numerical calculation of your grade on a quiz/test/exam, please bring it to the attention of the instructor immediately. This includes errors in adding up the total points lost and calculating the final grade. It does not include issues with the interpretation of your answers to the questions or the amount of partial credit given.

If you have an issue quiz/test/exam grading that is not just a numerical error, you must submit your request for regrading to the instructor in writing. Include your unaltered test paper and a brief description of why you believe that you should get additional points for your answer. Written requests for regrading a quiz/test/exam must be submitted within one week of the paper being handed back to you. Requests for regrading after that point – especially at the end of the semester, will not be considered.

You must keep up with the material and study for the quizzes, tests and final exam when they are scheduled. The grade for the class will be determined according to the assignments and percentages outlined above. No extra credit in the class will be given.

VCU Syllabus Statement: Students should visit here and review all syllabus statement information. The full university syllabus statement includes information on safety, registration, the VCU Honor Code, student conduct, withdrawal and more.

VCU Libraries: Use VCU Libraries to find and access library resources, spaces, technology and services that support and enhance all learning opportunities at the university. https://www.library.vcu.edu/